INCH-POUND

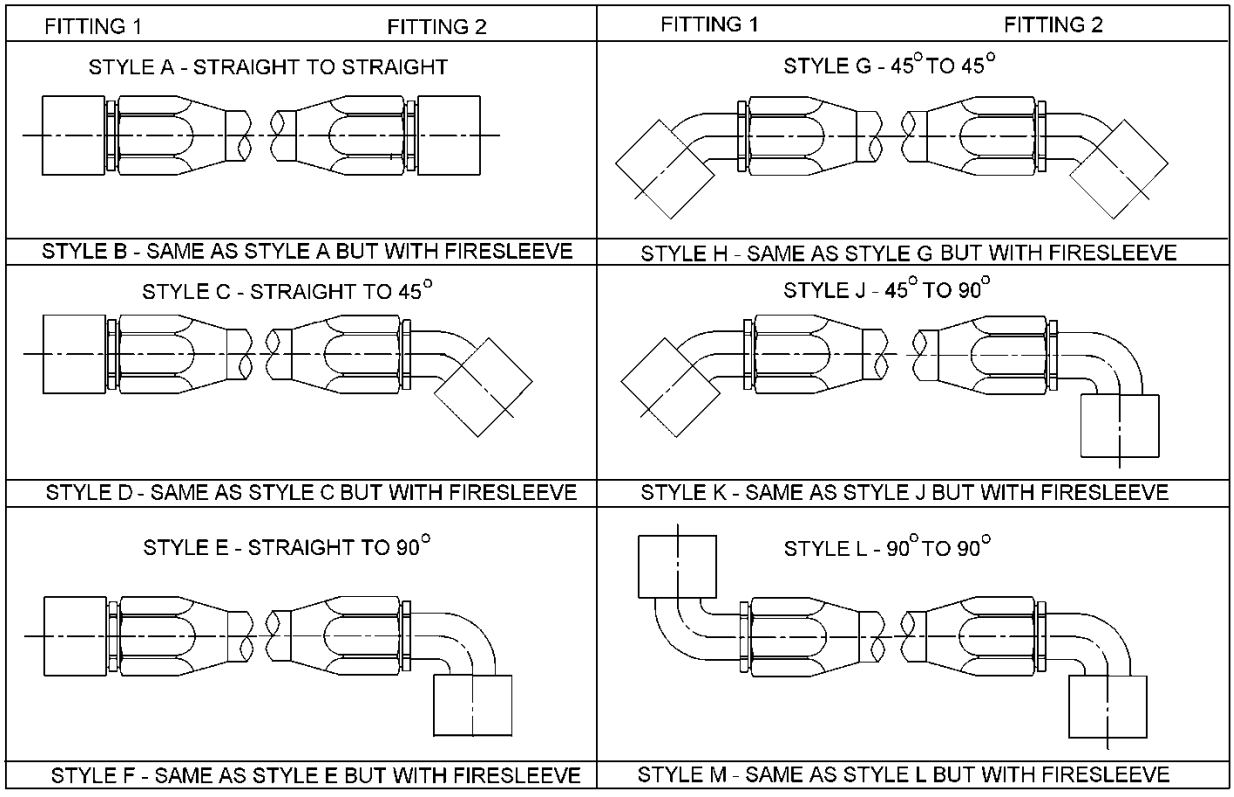
MIL-DTL-83796/11A
6 August 2014
SUPERSEDING
MIL-DTL-83796/11
11 August 2008

DETAIL SPECIFICATION SHEET

# HOSE ASSEMBLY, RUBBER, LIGHTWEIGHT, MEDIUM PRESSURE, INTERNAL SUPPORT COIL, FIELD ATTACHABLE END FITTINGS, FLARE TO FLARE, WITH LOCKWIRE HOLE

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83796.




NOTE: In the Part or Identifying Number (PIN) add the letter “L” after style for lockwire holes, example: EL.

FIGURE 1. Hose assembly styles, field attachable fittings, flare-to-flare.



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Hose assembly elbow fitting drop height. Elbow fitting drop height, A and B as shown on figure 2, are as specified in the applicable fitting specification sheet (see table I).

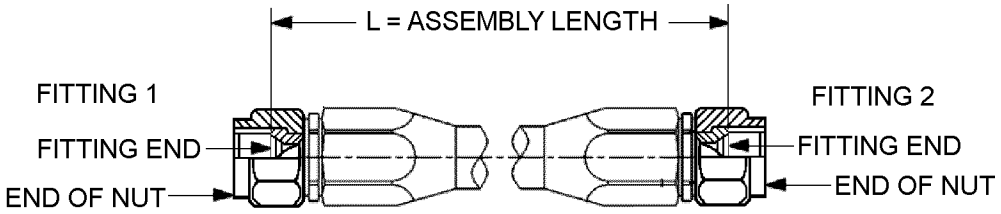
TABLE I. Hose assembly, field attachable fittings, HCOF, styles A thru M.

| Style | Fitting ends MIL-DTL-83798 | | MIL-DTL-83797 hose cut off factor (HCOF) 1/ inches verses hose dash size number | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Fitting 1 | Fitting 2 | -03 | -04 | -05 | -06 | -08 | -10 | -12 | -16 | -20 | -24 | -32 |
| A and B | M83798/1 | M83798/1 | 1.48 | 1.48 | 1.52 | 1.68 | 1.94 | 2.10 | 2.22 | 2.68 | 3.08 | 3.40 | 4.08 |
| C and D | M83798/1 | M83798/2 | 1.68 | 1.81 | 1.90 | 2.06 | 2.28 | 2.49 | 2.83 | 3.37 | 3.85 | 4.23 | 4.94 |
| E and F | M83798/1 | M83798/3 | 1.53 | 1.68 | 1.74 | 1.91 | 2.12 | 2.32 | 2.70 | 3.27 | 3.77 | 4.17 | 4.90 |
| G and H | M83798/2 | M83798/2 | 1.88 | 2.14 | 2.28 | 2.44 | 2.62 | 2.88 | 3.44 | 4.06 | 4.62 | 5.06 | 5.80 |
| J and K | M83798/2 | M83798/3 | 1.73 | 2.01 | 2.12 | 2.28 | 2.46 | 2.71 | 3.31 | 3.96 | 4.54 | 5.00 | 5.78 |
| L and M | M83798/3 | M83798/3 | 1.58 | 1.68 | 1.96 | 2.14 | 2.30 | 2.54 | 3.18 | 3.86 | 4.46 | 4.94 | 5.72 |

1/ Example of hose cut-off factor: for a -04, 1/4 hose OD, 18-1/2 assembly length, style D, the bulk hose length = 18.50 - the HCOF = 18.50 -1.81 = 16.69. Bulk hose length required is 16.69.

Hose assembly length. The hose assembly length is calculated as shown on figure 2.

Hose assembly elbow fitting drop height. Elbow fitting drop height B and C is calculated as shown on figure 2. Drop height dimensions for B and C can be found in the applicable MIL-DTL-83798 slash sheet as specified in table I herein.

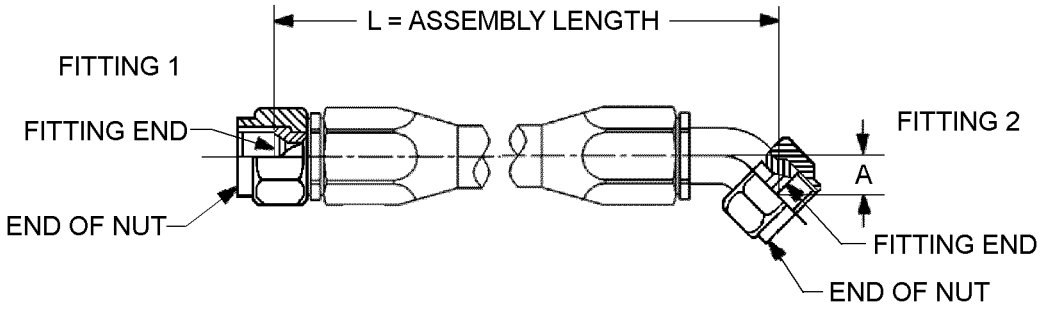



STYLE A or B

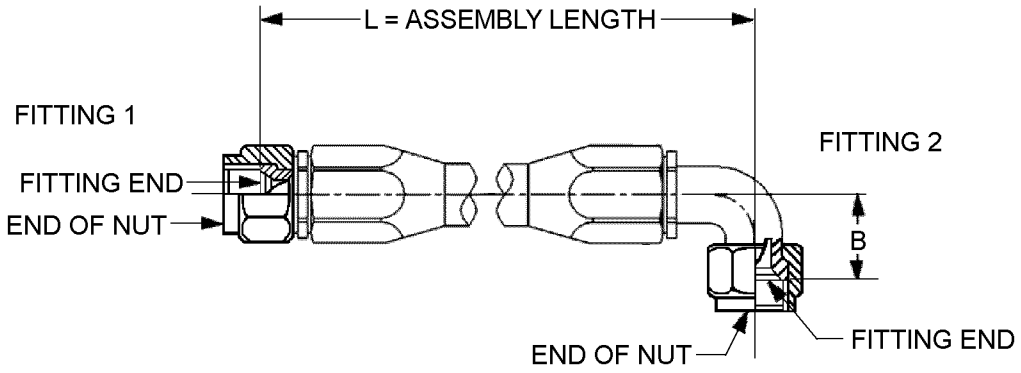
FIGURE 2. Hose assembly dimensions.



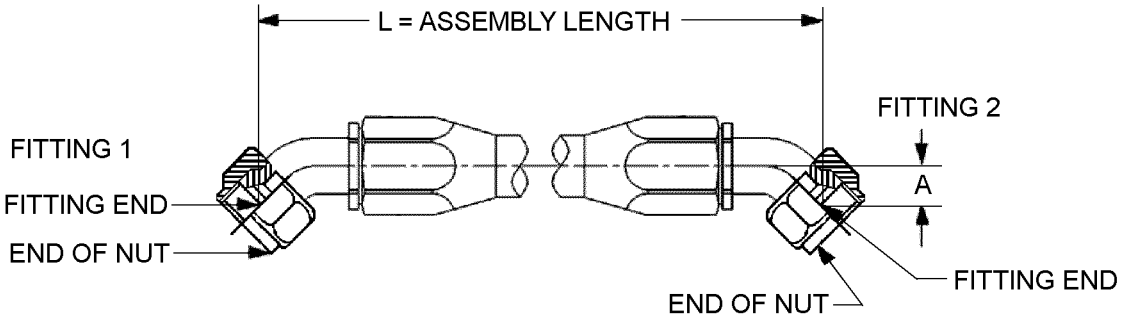
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STYLE C or D




STYLE E or F

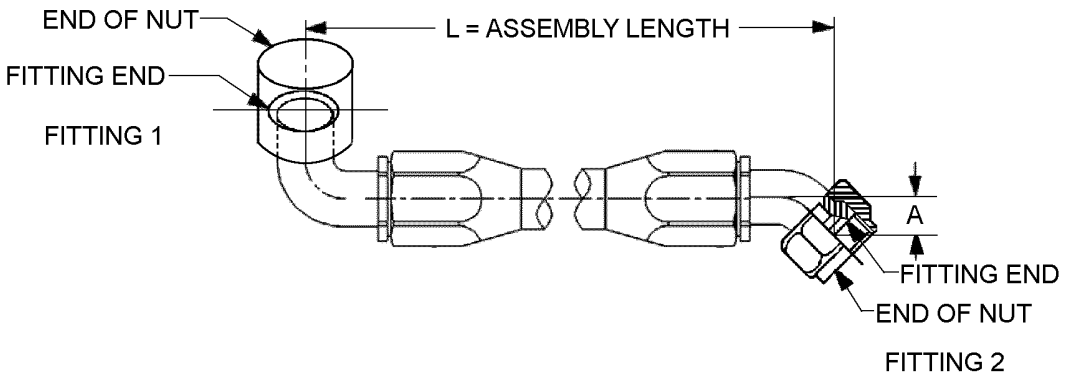



STYLE G or H

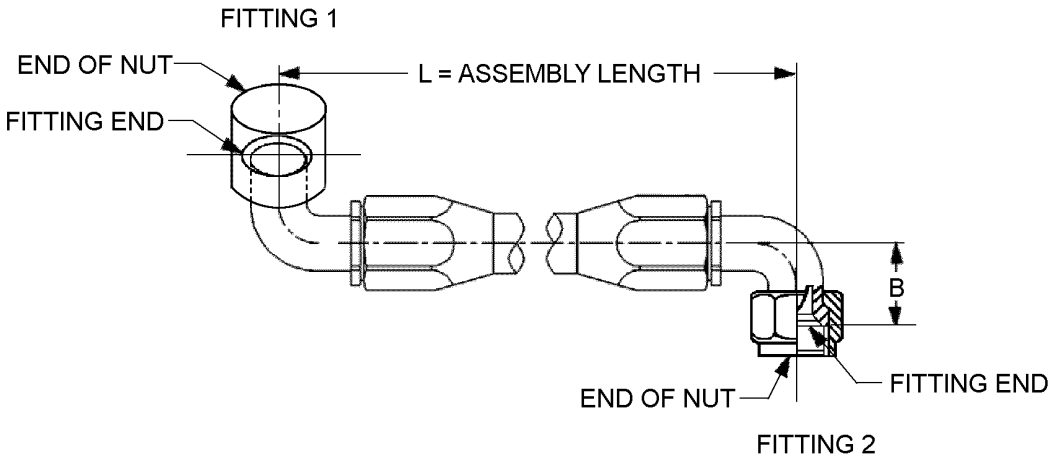
FIGURE 2. Hose assembly dimensions - Continued.



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STYLE J or K




STYLE L or M

NOTES:

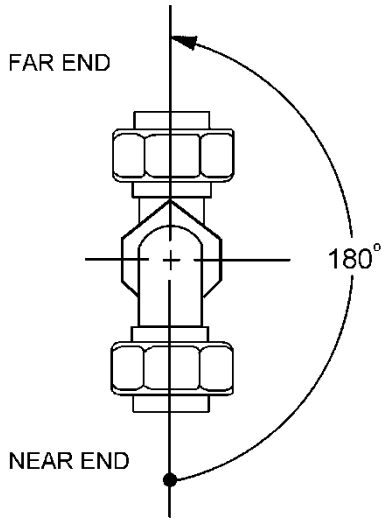
1. For straight hose assembly types the length “L” is to be measured between straight fitting end along a straight line parallel to hose, with hose laid out horizontally and straight, to straight fitting end.
2. For straight and 45° hose assembly types the length “L” is to be measured between sealing surface of fitting end along a straight line parallel to hose, with hose laid out horizontally and straight, to center line of 45° fitting end.
3. For straight and 90° hose assembly types the length “L” is to be measured between straight fitting end along a straight line parallel to hose, with hose laid out horizontally and straight, to center line of 90° fitting.
4. 45° and 90° hose assembly types the length “L” is to be measured between center line of 45° fitting end along a straight line parallel to hose, with hose laid out horizontally and straight, to center line of 90° fitting.
5. 90° hose assembly types the length “L” is to be measured between center line of 90° fitting end along a straight line parallel to hose, with hose laid out horizontally and straight, to center line of 90° fitting end.

FIGURE 2. Hose assembly dimensions - Continued.



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Angular displacement. Angular displacement shall be as shown on figure 3.




NOTES:

1. Angular displacement for hose assemblies with elbow fittings on both ends shall have the angular displacement between elbows, measured counter-clockwise from the centerline of the nearest fitting, positioned at six-o' clock, to the centerline of the other fitting (see SAE-J517).
2. The near end of the connector shall be put in numerical order relative to the far end (see table I). Example: Near end adapter 45°, far end 90°.
3. Angular displacement shall be measured in degrees with a tolerance of ± 3° for lengths up to 2 feet (61 cm) and ± 5° for all lengths over 2 feet (61 cm).
4. The angular displacement shall be expressed in two digits number of degrees divided by 5, example 45/5 = 09.
5. Making the angular determination in the wrong direction will result in an unacceptable part.

FIGURE 3. Elbow fitting drop height and angular alignment.



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REQUIREMENTS

Hose assemblies shall be specified on figures 1, 2, and 3.

Fittings shall mate with parts in accordance with SAE-AS4395.

Hose assembly length tolerance shall be in accordance with MIL-DTL-83796.

Internal support coil. Internal support coil shall be in accordance with DLA Land and Maritime drawing 08001.

Support coil length. The support coil length shall be the overall hose assembly length minus the sum of the overall lengths of each end fitting.

Size codes shall be as specified in table II.

TABLE II. Size code. 1/ 2/

| Dash number | | -03 | -04 | -05 | -06 | -08 | -10 | -12 | -16 |
|---|---|---|---|---|---|---|---|---|---|
| Size code | | B | E | F | G | H | J | K | M |
| Hose OD reference | Inches fraction | 3/16 | 1/4 | 5/16 | 3/8 | 1/2 | 5/8 | 3/4 | 1 |
| | Inches decimal | .188 | .250 | .313 | .375 | .500 | .625 | .750 | 1.000 |
| | mm | 4.78 | 6.35 | 7.95 | 9.53 | 12.70 | 15.88 | 19.05 | 25.40 |

| Dash number | | -20 | -24 | -32 |
|---|---|---|---|---|
| Size code | | N | P | R |
| Hose OD reference | Inches fraction | 1 1/4 | 1 1/2 | 2 |
| | Inches decimal | 1.250 | 1.500 | 2.000 |
| | mm | 31.75 | 38.10 | 50.80 |

1/ Dimensions are in inches.
2/ Metric equivalents are given for information only.

Materials and finishes for hose fittings shall be in accordance with MIL-DTL-83798 and table III.



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TABLE III. Nut and nipple material and finish code. 1/

| Material and finish code | Material | Plating Finish |
|---|---|---|
| Blank | Steel (sizes -3, -4, and -5) | Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2. |
| | Aluminum (sizes -6 and above) | Anodize in accordance with MIL-A-8625, type II |
| A | Steel (sizes -3, -4, and -5) | Aluminum-nickel in accordance with ASTM F1136/F1136M, grade 3, NC |
| B | Aluminum (sizes -6 and above) | Anodize above with NAVAIR trivalent chromium pretreatment (TCP) in accordance with MIL-DTL-81706, type II, class 1A. |
| CN | Steel (sizes -3, -4, and -5) | Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2 with NAVAIR TCP in accordance with MIL-DTL-81706, type II, class 1A. |
| H | Steel (sizes -3, -4, and -5) | Zinc phosphate finish in accordance MIL-DTL-16232 type Z, class1 2/ |
| J | Steel (sizes -3, -4, and -5) | Zinc plating in accordance with ASTM B633; type II or III, Fe/Zn 5, or ASTM B695, type II, class 5 |
| N | Steel (sizes -3, -4, and -5) | NAVAIR TCP above. |
| R | Steel (sizes -3, -4, and -5) | Zinc plating in accordance with ASTM B633; type VI, Fe/Zn 5. |
| S | Corrosion resistant steel | No additional finish. Passivation in accordance with SAE-AMS2700, method 2 |
| T 3/ | Titanium (all sizes) | Annealed |
| V | Steel (sizes -3, -4, and -5) | Zinc-nickel in accordance with SAE-AMS2417, type 1 |
| Z | Zinc any type above | PIN code F, G, H, J, R, V |
| ZN | Zinc any type above | PIN code F, G, H, J, R, V with NAVAIR TCP in accordance with MIL-DTL-81706, type II, class 1A. |

1/ Shall be capable of withstanding minimum of 96 hours salt spray.
2/ Hexavalent chromium free.
3/ Titanium shall not be used in oxygen systems.

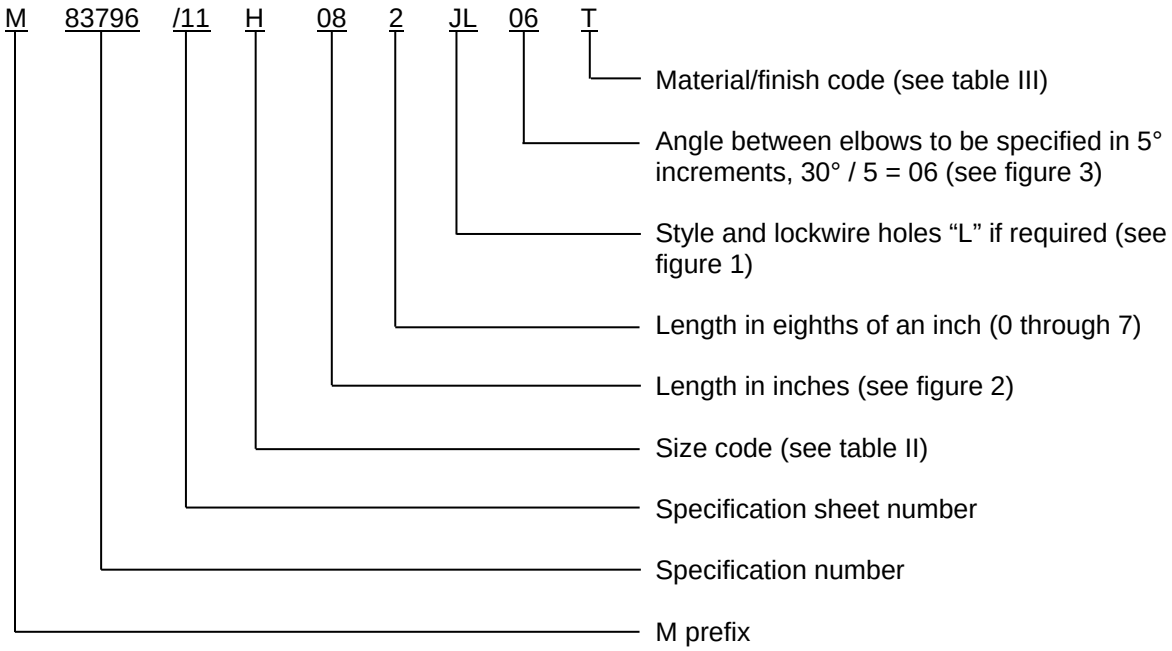
Trivalent wrenchability. When the finish has been damaged due to poor wrenchability, the surface of the connector shall be touched up using the brush plating process below. The term “trivalent wrenchability” is used to evaluate the ability of the finish to withstand abrasion from an excessive amount of wrenching.

a. Brush plating of hard chromium by electrodeposition shall be in accordance with SAE-AMS2451/5.
b. Brush plating of medium-hardness, low stress nickel by electrodeposition shall be in accordance with SAE-AMS2451/9.
c. Brush plating of NAVAIR TCP shall be in accordance with MIL-DTL-81706, type II, class 1A, material form 1 through 6, application method B. Example of a PIN: M817062A6B.



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PIN example:




M83796/11H082J06T indicates - Hose assembly for ½ hose OD, 8 1/4 inches in length, style J, 30° between elbows, titanium alloy.

M83796/11H082JL06S indicates - Hose assembly for ½ hose OD, 8 1/4 inches in length, style J with lockwire holes, 30° between elbows, corrosion resistant steel.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-83796, this document references the following:

| | | |
|---|---|---|
| MIL-A-8625 | ASTM B633 | SAE-AMS2451/5 |
| MIL-DTL-16232 | ASTM B695 | SAE-AMS2451/9 |
| MIL-DTL-81706 | ASTM F1136/F1136M | SAE-AMS2700 |
| MIL-DTL-83797 | SAE-AMS-C-81562 | SAE-AS4395 |
| MIL-DTL-83798 | SAE-AMS-QQ-P-416 | SAE-J517 |
| 08001 | SAE-AMS2417 | |



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CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4720-2014-027)

Review activities:
Army - AT
Navy - MC, SA, SH
Air Force 71, 85

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